

Lecture 11

- Differentiate between various media access methods
- Distinguish between CSMA/CD, CSMA/CA, CSMA/CA RTS/CTS, and CSMA/CA PCF
- Explain CSMA/CD, CSMA/CA, and CSMA/CA RTS/CTS
- Describe the background and function of flow control
- List examples of protocols for flow control
- Explain the implementation of logical to physical addresses using ARP
- Identify the contents of an ARP message
- State the tasks of the network layer
- Explain the tasks of a router
- Present the tasks of a Layer 3 switch
- Name the differences and similarities between a Layer 3 switch and a router
- Identify the tasks of a gateway and its synonyms
- Illustrate broadcast and collision domains
- Distinguish addressing types in the network layer

Data Link Layer:

Main Purpose: To provide reliable and error-free communication between two directly connected devices on the same network, by encapsulating network-layer data into frames, adding physical addresses (MAC), detecting/correcting errors, and controlling access to the shared medium.

N.B: when we say 2 already connected devices we mean that the session layer has already opened and maintained a session and the 2 devices (e.g: laptop, server) are connected.

“Packet shaping from data of Network Layer” means that the Data Link Layer takes raw packets from the Network Layer and prepares them as properly formatted frames with addressing and error-checking information so they can travel over the physical network.

On the receiving of physical bits from the physical layer the data link layer do something called **deframing**: means the Data Link Layer takes the raw bitstream from the Physical

Layer, identifies complete frames, checks for errors, and extracts the original Network Layer packet for further processing.

Error detection by Checksum while transmitting the data; after the data link layer have added the mac address for the destination and the source, the ip packet it calculates how long it should take for this package to reach the other end, and adds this to the tail of the frame. The receiving data link layer calculates the time taken by the package and compare it against the real number if they are not equal (error detection), the receiving asks the sender to send the packet again (correction).

“Provision of physical addresses” means the Data Link Layer assigns and uses MAC addresses to identify devices on the same network segment and deliver frames to the correct recipient. It translates the IP address of the receiver into its mac address and put it in the head of the packet.

Access control according to media access schemes; what that means is that the data link layer is responsible for making sure that the data does not collide on the physical layer using protocols like CSMA/CD CSMA/CD.

Typical protocols: Ethernet, WLAN, Bluetooth, PPP: these protocols have CSMA/CD CSMA/CA, MAC Addressing and other protocols inside of them.

Devices working (only) in this layer: Bridge, Layer-2-Switch (Multiport Bridge), Modem.

Devices: Bridge, Layer-2-Switch:

Bridge: connects 2 networks (typically with different transmission media). It has 2 ports now each port becomes a collision domain. It connects two separate network segments and allows devices on both sides to communicate.

Layer-2 Switch (aka multiport bridge) connects more than 2 networks: A Layer-2 switch is essentially a bridge with many ports. Each port connects to a different network segment or device. It forwards frames based on MAC addresses. And each port (device) becomes a collision domain.

Bridges and layer-2 switches have a purpose to increase the ethernet distance span, since the ethernet cable can be at maximum of 100m.

What bridges / Layer-2 switches actually do: Learn MAC addresses. Filter frames (decide where to send them). Forward frames only to the correct port

By bridge/layer-2-switch separated networks separate collision domains: when using a bridge or a layer 2 switch each port becomes its own collision domain and each attached end device becomes its own collision domain.

layer 2 switch and a bridge actually do some traffic controlling not just blind broadcasting to all

on the other hand hubs and repeaters: those just receive and broadcast signals to all connected media which makes hubs and repeaters a single collision point where the device itself is the collision domain not the connected devices.

A bridge/layer 2 switch are not collision domain, no collision can happen on them but rather the surrounding domains. A hub is a collision domain.

MAC Address (Media Access Control):

What is a MAC Address: it's a unique identification number mapped on the hardware for NIC that the data link layer uses to help with network access control and frame delivery.

Why do we have MAC Addresses: it allows the data link layer to directly send data packets to certain devices on LANs, it helps switches to redirect the packet to the correct destination. It helps with the CSMA/CD and other error detection and prevention protocols.

Data frames with a destination MAC of **ff:ff:ff:ff:ff:ff** are meant to reach (be broadcasted) to all devices on the same LAN

MAC Addresses are unique and not changeable.

It consists of 6 bytes each byte separated by a coma, 6 bytes = 48 bits 2^{48} means we have trillions of options.

MAC Addresses Uniqueness:

Managed by IEEE → They assign MAC address blocks to companies.

MAC Address Blocks: Small (MA-S), Medium (MA-M), Large (MA-L)

MA-L / OUI: Companies get **first 24 bits** to identify themselves (Organizationally Unique Identifier)

Last 24 bits: Each company assigns **unique IDs to their devices** (NICs)

Example OUI:

- 90:8D:6E → Dell
- 00:04:0F → Asus
- D4:F3:2D → Intel

Multicast, normal, and local MAC Addresses:

A multicast MAC address is a MAC Address that is meant for a group of end devices.

A normal MAC Address is a one that is globally unique for one end device.

Local MAC Address is a MAC address is MAC address made by the software to be reserved for a VM.

Reading A MAC Address: the first one byte is our main focus, we look at the first 2 bits of the first byte

The first bit of the first byte if its 0 then it's a unicast (meant for a single unique MAC address) if its 1 it's a multicasting (meant for a group of end devices)

The second bit of the first byte if its 0 its globally unique MAC address if its 1 then its locally done (vm)

Requires a „Mapping“ Protocol to match logical and physical addresses, e.g. in Ethernet Address Resolution Protocol (ARP) or Neighbor Discovery Protocol (NDP). Those take place between layer 3 and layer 2. Between transport and data link layer.

Security Aspects of MAC Addresses:

MAC filters are often used

- Routers can allow or block devices based on MAC addresses.

MAC addresses are visible

- Any device on the network can see other devices' MAC addresses.

Low security

- MAC addresses can be changed in software, so filtering alone is not very secure.
- Example: `ip link set dev <Interface> address <MAC>`

MAC spoofing threat

- Attackers can pretend to be another device by changing their MAC, especially in WiFi networks.

Reasons for Media Access Control:

Since we have multiple end devices many times these end devices share the same media, making it very possible of a single collision domain creation. In normal situation or in the past we had no coordination which is a huge problem.

A solution for this is called Media access control schemes: its goal is Minimizing data re-transmissions to minimize waiting time (increased latency) by ensuring that collision shall not happen and in case it happened that the data shall be transmitted again.

This is achieved through CSMA/CA & CSMA/CD

CA → Collision Avoidance used in WLAN

CD→Collision detection Used in Ethernet

TDMA → Time division multiple access (similar to round robin in OS)

Time Division Multiple Access (TDMA):

TDMA is a channel access method used in networks where multiple devices share the same frequency channel, but avoid collisions by dividing time into slots. It works the same as round robin where each end device gets a time slot to send data frames ones at a time.

Each frame at a given time; TDMA splits the communication channel into time slots where many devices share the same frequency. Each device gets its own time slot where it can only use the channel to send data packets at this time. This prevents data collision

Required: Each terminal „knows“ the exact time to send its messages; this means that the protocol assigns a certain known time for each end device to start transmitting its data on the media. Works like round robin in some sense

Exact time distribution; that means that the TDMA protocol ensures that the access point distributes the time slots of all the connected end devices so these end devices know when their time slots begins and ends and when the other devices will begin and end this synchronization prevents collision.

GSM (Global System for Mobile Communications): 2G cellular standard, Uses TDMA to allow multiple users to share the same frequency channel

UMTS (Universal Mobile Telecommunications System): 3G cellular standard, Uses TDMA combined with CDMA (called WCDMA for wideband CDMA)

TDMA is a data link layer that is implemented to allow multiple access channel

CSMA/CD:

CSMA/CD is a media sharing access control, its inside the data link layer. It allows multi end devices to use the same media without collision of the signals.

The main idea (usually implemented in bus networks) is that an end device in a shared media will continuously check if the media (cable) is idle and if so it will start transmitting data and check for collision if no collision has happened it ends. If a collision happened, it sends a jam signal (a signal to all end devices to tell them to stop sending packets) then it waits randomly if the number of maximum attempts is not reached, if the number of maximum attempts was reached it then tells the upper layer. After it waited for random amount of time it again checks if the media is idle.

Back-off time: Random waiting time; when a signal collides the 2 transmitting devices have a random back off timing before they check the media again.

Minimum value $< t <$ maximum value of Contention Window alue (CW): this time should be somewhere between a minimum value and a maximum value called CW, where this CW growth exponentially (doubles each time a collision happen)

Maximum numbers of transmission attempts ≤ 16 : for a single packet we have maximum of attempts up to 16 if we exceeded 16 or 16 collisions has happened for the same data frame the data link layer communicates with the upper layers and does not try no moe.

Bus topology: this protocol is very important for bus topology since in bus topology all end devices share the same physical media. But its rarely used.

CSMA/CA Collision Avoidance:

CSMA/CA is a media access control method mainly used in wireless networks where collisions cannot be detected. Instead of detecting collisions, devices try to avoid them in advance by waiting, listening, and coordinating before sending data.

Node sends a frame after a Distributed Interframe Space (DIFS) time period; before sending the data, and after getting the signal or making sure that the media is idle to send data. The end device waits a fixed amount of time called DIFS as to make sure that the media is totally idle and no collision can happen.

Receiver acknowledges its reception after a Short Interframe Spacing (SIFS) time period; after receiving a data frame the receiver waits a fixed time called SIFS before it sends an acknowledgement of the data reception. This also ensures collision is avoided. SIFS are less than DIFS which means it has higher priority.

Other terminals have to wait sending for a DIFS and random backoff time; that means that the backoff time only happens if the CSMA/CA have many devices that want to send data at the same time, they have to wait the fixed DIFS and also random backoff time

Hidden Terminal; A device that cannot hear a certain device, then it's hidden or the current device cannot communicate with it or know when it's sending or receiving data. However, they both can see the access point AP.

RTS/CTS solution for hidden terminals; RTS request to send, CTS clear to send. The first one lets the end device know that it wants to send the data frame the CTS lets the sending device know that the media is idle to receive. Now when a 3rd hidden terminal checks for media and hears from the access point that the terminal is in clear to send state it does not send anything it knows that other device is sending.

Some real life RCT/CTS CSMA/CA applications: WIFI (802.11), Apple Talk • Controller Area Network (CAN)

Pre messages: the RTS/CTS messages are called pre messages (dataframe). For RTS this is only going to the receiver the AP. However, the CTS is sent to all nodes.

This RTS/CTS only takes place when an end point wants to send data to the access point, the access point does not send RTS only CTS

Disadvantages: since RTS/CTS adds more steps it makes the communication take more latency.

Point Coordination Function (PCF):

Access Point coordinates access to the transmission media

All terminals must register to AP

AP asks terminals to transmit

Rarely implemented

Address Resolution Protocol (ARP):

The network layer (internet layer) layer 3, works with IP addresses while the layer 2, data link layer works with MAC addresses. The ARP is something that maps the known IP address to an unknown MAC address it sits between these 2 layers.

And ARP request is sent to the known IP address asking for the MAC address of the corresponding IP address and providing the requester's MAC address.

In case this is a device outside the network gateway, the device communicates with the IP and MAC address of the router

An ARP request is broadcasted to all the end devices on the same network

It sends the ARPS request to ff:ff:ff:ff:ff:ff

IP address and MAC address are stored in ARP cache of the requestor.

To run this ARP cache we write the command `arp -n`

ARP Cache:

The ARP cache is a temporary storage table on a device (PC, router, switch) that keeps track of IP addresses and their corresponding MAC addresses.

Goal: Speed up network communication by avoiding repeated ARP requests. Once a MAC address is resolved, it is stored locally in the cache. Next time, the device can look it up instantly → much faster. Instead of sending multiple requests to get the same mac address we already had before.

The ARP Cache table has:

IP protocol version (ipv4, ipv6)	IP address	MAC address	TTL
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TTL: time to live, is the time the MAC address is going to be stored in the cache before its removed. Which is usually 5 minutes

Broadcast Domains:

A broadcast domain is a network where when a data frame is sent by one end device its is broadcasted (forwarded) to all devices on this network.

This broadcast carrying a message then reaches all the connected devices.

Router/Layer-3-Switch separate broadcast domains. So each network connected to a layer-3 router becomes its own broadcast domain.

Devices on the Network- internet Layer -3 All separates broadcast domain

Devices on the Network Layer (Layer 3) are responsible for routing packets between different networks, using logical addresses (like IP).

The main devices are routers and Layer-3 switches, and sometimes gateways

Routers: Routers can connect local networks to wide area networks (WANs), such as the Internet. They handle different types of networks, often with NAT (Network Address Translation) to map local IPs to public IPs. They also forward packets based on IP addresses

Layer-3 switches: Forward packets based on logical addresses (IP addresses) rather than MAC addresses.

Gateway: Gateways translate between different protocols. Can work at Network Layer (3) or Transport Layer (4).

Default Gateway: is the device your LAN computers send packets to when the destination is outside the local network.